Fiscal Year:	FY 2016	**Task Last Updated:**	FY 04/27/2016
PI Name:	Otto, Christian M.D.		
Project Title:	Prospective Observational Study of Ocular Health in ISS Crews		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	02/03/2012	**End Date:**	01/05/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 1/5/2017 as PI moved in January 2017, per A. Allcorn/HRP (Ed., 1/20/17) NOTE: End date change to 11/16/2017 per R. Brady/HRP (Ed., 5/2/14) NOTE: Title change implemented per HRP Element; previously "International Space Station (ISS) Ocular Surveillance Protocol" (Ed., 1/27/14)		
Key Personnel Changes/Previous PI:	JANUARY 2017: Principal Investigator Dr. Christian Otto moved and new PI is Brandon Macias, Ph.D.		
COI Name (Institution):	Gibson, C. Robert O.D. (Coastal Eye Associates--Texas) Sargsyan, Ashot M.D. (Wyle Integrated Science and Engineering) Alexander, David M.D. (NASA Johnson Space Center) Ploutz-Snyder, Robert Ph.D. (Universities Space Research Association) Riascos, Roy M.D. (University of Texas Medical Branch) Patel, Nimesh O.D., Ph.D. (University of Houston) Samuels, Brian M.D., Ph.D. (University of Alabama at Birmingham) Kramer, Larry M.D. (The University of Texas) Lee, Stuart Ph.D. (Wyle Integrated Science and Engineering)		
Grant/Contract No.:	Directed Research		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The International Space Station (ISS) Ocular Surveillance Protocol aims to systematically gather physiological data to characterize the Risk of Microgravity-Induced Visual Impairment/Intracranial Pressure (VIIP) on crewmembers assigned to a 6 month ISS increment. The data collected will mirror Medical Requirements Integration Documents (MRID) requirements and testing performed during annual medical exams. The frequency of in-flight and postflight testing will be increased to more accurately assess changes that occur in the visual, vascular, and central nervous systems upon exposure to microgravity and induction of fluid shifting. Monitoring in-flight changes, in addition to postflight recovery, is the main focus of this protocol. A data sharing plan with Medical Operations will reduce redundancy of data acquisition. Preflight, in-flight, and postflight measures include: tonometry, ocular ultrasound, fundoscopy, and visual acuity; while magnetic resonance imaging (MRI), optical coherence tomography (OCT), and bio-microscopy will be captured preflight and postflight exclusively. Two additional, non-MRID measures, blood pressure and cardiac output, will be collected preflight, in-flight, and postflight to assess vascular compliance. Data collection will begin one year prior to flight, continue in-flight approximately every 30 days, and through to one year postflight. In circumstances where abnormalities may persist beyond one year, postflight data will continue to be collected, but as per the MRID requirements (MedB 1.10) and VIIP clinical practice guidelines. Expected Outcomes 1. It is expected that some crewmembers will experience meaningful and detectable in-flight changes in at least one or more of the following: visual acuity, intraocular pressure, optic disc edema (papilledema), chorioretinal folds, optic nerve sheath distention, optic nerve tortuosity, optic nerve-to-sheath ratio, globe flattening, and retinal “cotton-wool spots.” 2. It is expected that some crewmembers will experience meaningful pre- to postflight changes in one or more of the following: visual acuity, intraocular pressure, optic disc edema (papilledema), chorioretinal folds, optic nerve sheath distention, optic nerve tortuosity, optic nerve-to-sheath ratio, globe flattening, retinal “cotton-wool spots”, vascular compliance, CSF (cerebral spinal fluid) velocity through the aqueduct of Sylvius, and retinal nerve fiber layer. 3. It is expected that if an in-flight or postflight measure deviates from preflight baseline measures, it may have a prolonged recovery to baseline (preflight values) that is positively associated with severity. Specific Aims For all measured variables and using currently available on-orbit methodologies, this study has the following specific aims: 1. Characterize the nature of in-flight visual, vascular, and central nervous system changes during six months exposure to microgravity. 2. Document changes from pre- to postflight. 3. Delineate the interaction between individual susceptibility and severity of symptoms. 4. If changes occur, establish the postflight time course for recovery to baseline. See also: http://www.nasa.gov/
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research and there is insufficient time. This research is highly constrained because it is proposing additional data collections of MRID measures pre-, in-, and postflight. Since the co-investigative team collects the MRID data, they are the best source to collect this data as well. Due to the visibility of this risk, there is pressure to characterize the visual changes associated with spaceflight in order to begin to identify the underlying cause. The results of this study will help define and frame the new Risk of Microgravity-Induced Visual Impairment/Intracranial Pressure.
Research Impact/Earth Benefits:	Terrestrial Benefits 1. The VIIP syndrome has similarities to terrestrial medical conditions such as glaucoma, Normal Pressure Hydrocephalus (NPH), Idiopathic Intracranial Hypertension (IIH), and high-altitude related illnesses 2. Advances in the tools, techniques, and countermeasures that NASA develops in its VIIP research will benefit these terrestrial clinical populations 3. Identifying the cause(s) and risk factors for the VIIP syndrome will also inform the cause(s) and risk factors for these terrestrial conditions.
Task Progress:	The full complement of 13 subjects has been recruited to the Ocular Health study. To date, six subjects have completed all phases of data collection--preflight, inflight and postflight. Three subjects are in postflight data collection having completed preflight and inflight collection, two subjects are currently inflight, and two subjects are in the preflight data collection phase. Of the subjects who have completed the inflight phase, one has been identified as a VIIP case with clinical grade disc edema (swelling of the optic nerve where it enters the posterior portion of the eye). The VIIP case stands out from the eight non-cases as distinct in the majority of measured parameters. Data to date indicate that optic nerve sheath subarachnoid space distension (a fluid filled space that surrounds the optic nerve as it traverses from just inside the skull to the eyeball) and eye globe compression dominate the clinical picture and are positively associated with the degree of retinal nerve fiber layer thickening, also referred to as optic disc edema. In addition, optic nerve distension may be associated with vascular changes recorded in the central retinal artery and vein which supply and return blood from the retina. Lastly, evidence to date suggests that intraocular pressure may be

	playing a role in modulating the degree of optic disc edema. NOTE: JANUARY 2017: Principal Investigator Dr. Christian Otto moved and new PI is Brandon Macias, Ph.D. See that project with same title for continuation information and reporting.
Bibliography Type:	Description: (Last Updated: 04/28/2016)
Abstracts for Journals and Proceedings	Otto C, Barr Y, Gibson C, Alexander D, Sargsyan A, Ploutz-Snyder R, Brady R, Riascos R, Patel N, Samuels B, Garcia K, Kramer L, Alexander D, Lee S. "The Prospective Observational Study of Ocular Health in International Space Station (ISS) Astronauts: The Visual Impairment Intracranial Pressure Risk." 25th Annual American Glaucoma Society Meeting, Hotel del Coronado, Coronado, CA, February 26-March 1, 2015. 25th Annual American Glaucoma Society Proceedings. 2015 Feb. p. 80. , Feb-2015
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